AI Engineer

Step 5: Learn NLP & Computer Vision

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

NLP and computer vision are application domains rather than single algorithms. Modern NLP often uses transformer architectures and pretrained models; computer vision combines image processing with learned models such as CNNs and modern vision architectures.

Hugging Face's current course covers Transformers, Datasets, Tokenizers and Accelerate, while also emphasizing traditional NLP foundations. ■cite■turn0search12■

Core Concepts — Detailed Breakdown

- NLP: text normalization, tokenization, vocabulary/subwords, embeddings and sequence representations.
- Text tasks: classification, named entity recognition, question answering and generation.
- Transformers: attention, contextual representations and pretrained/fine-tuned models.
- Computer vision: pixels/tensors, resizing, normalization and augmentation.
- CNN intuition: local receptive fields and learned filters.
- Vision tasks: classification, object detection and segmentation.
- Transfer learning: start from a pretrained representation when data/compute are limited.
- Evaluation: choose metrics appropriate to task and class balance.

What You Should Be Able To Do

- Build a simple text classifier.
- Inspect tokenization.
- Fine-tune or use a pretrained model for a small task.
- Build an image classifier.
- Evaluate errors by class and inspect representative failures.

Hands-on Project / Practice

- Choose either sentiment/text classification or image classification. Use a pretrained model where appropriate, compare it with a simple baseline, and document failure cases.

Recommended References

- Hugging Face Course — https://huggingface.co/docs/course/
- TensorFlow Learn — https://www.tensorflow.org/learn
- PyTorch Tutorials — https://pytorch.org/tutorials/
- NLP Tutorial with Python & NLTK — https://www.youtube.com/watch?v=X2vAabgKiuM
- OpenCV Python Course — https://www.youtube.com/watch?v=P4Z8_qe2Cu0

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.